

FINAL REPORT ? AB-Cloud v19 ? full 37-test verification suite

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What this folder is

FINAL_REPORT is the LAST folder of the run and gathers ALL run information: the master report (this document, in md/html/pdf), COPIES of every test's report files (reports/), CO

How to reproduce

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Single test: `julia ab cloud v19.jl --test 33 --no-two-pass`

Disorder cap: `--ab-w-max 1.0` (Tests 32/33/36 use $W_{\text{eff}} = \min(ab \cdot W, W_{\text{max}})$)

Test 36 targets: --ab37-n-eigs-pass1 25000 --ab37-n-eigs-pass2 50000

Labs: main menu 'l' (Physics Lab), 'd3' (3D Lab); every experiment writes its own report into the same run folder.

```
zeros_count      = 50000 (source: auto)
```

Configuration snapshot (end-of-run values)

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lattice secondary = 96x96 Nv=[2] (two-pass=true, primary-only=false)

third pass (SERIES) = true (Tests 19/29/31 ? 19c/29c/31c real report series)

```
alpha      = ab_alpha=0.5, ab_alpha_test16=0.5
```

disorder = ab W=4, ab W max=1 ? W eff(Tests 32/33/36)=1.00

test33 comparison row = W cmp=0.50, Nv cmp=144

realizations = ab n_realizations=5, center fraction=0.6

```
byte robustness      = tol=0.02, n_bytes=200000, seed=12345, T37 targets: pass1=25000 / pass2=50000 eigs
```

lab = L=72, ?=0.5, W=4, q=1, Nv=4, random/torus/monumental, seed=12345

```
3d = L=10, ?=0.5, tz=1, W=0, mass=0, boundary=:torus, lines=0 q=1, seed=777
```

```
reporting = enabled=true, dir=results, dpi=600, formats=md,html,pdf,docx
```

Test 1 [PASS] bN_convergence ? Test 1: b(N=50000) = 1.2126 ? PASS

Test 1 [PASS] bN convergence ? Test 1 [HARDCORE pass 2]: 8 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 2 [PASS] bN_monotonicity ? Test 2: Monotonicity violations: 0 / 499 ? PASS

Test 2. [PASS] bN monotonicity ? Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Verdicts ? 77 tests: 66 PASS, 0 WARN, 11 FAIL, 0 DONE

Test 3 [PASS] bN rate ? Test 3: ? = 0.1685 (empirical; no expected value), R?=0.9895 ? PASS

Test 3 [PASS] bN rate ? Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 4 [FAIL] gue_ks_full ? Test 4: KS D=0.0216, p=0.0 ? WARN

Test 4 [PASS] gue_ks_full ? Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 5 [FAIL] gue_ks_highT ? Test 5: Best high-T p-value=0.0 ? WARN

Test 5 [PASS] gue_ks_highT ? Test 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 6 [FAIL] chi2_hist ? Test 6: ?? p=0.0 ? WARN

Test 6 [PASS] chi2_hist ? Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 7 [PASS] decay_slope ? Test 7: Slope=-0.1504 (empirical), 95%CI=[-0.1594,-0.1414] ? PASS

Test 7 [PASS] decay_slope ? Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 8 [PASS] residuals ? Test 8: Runs=3 (exp=5.8) ? PASS

Test 8 [PASS] residuals ? Test 8 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 9 [PASS] bootstrap_ci ? Test 9: Bootstrap slope=-0.1746 (empirical), 95%CI=[-0.1895,-0.1552] ? PASS

Test 9 [PASS] bootstrap_ci ? Test 9 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 10 [PASS] cross_validation ? Test 10: Max deviation=8.5% ? PASS

Test 10 [PASS] cross_validation ? Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Test 11 [FAIL] anderson_darling ? Test 11: A?=61.4924, MC p=0.0000 ? WARN (best high-T subrange T>31351 also rejects, p=0.0000; see height-resolved table)

Test 11 [PASS] anderson_darling ? Test 11 [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Test 12 [FAIL] two_sample_ks ? Test 12: 2-sample KS D=0.0233, p=0.010547 ? WARN

Test 12 [PASS] two_sample_ks ? Test 12 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 13 [PASS] number_variance ? Test 13: data closer to GUE in 9/9 L-values ? PASS

Test 13 [PASS] number_variance ? Test 13 [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 14 [PASS] spectral_rigidity ? Test 14: data closer to GUE in 9/9 L-values ? PASS

Test 14 [PASS] spectral_rigidity ? Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 15 [PASS] ab_construction ? Test 15: Hermiticity=OK, ?_TRB=0.1037, flux_v1=-0.0 [exp -0.0], flux_empty=0.0 ? PASS

Test 15 [PASS] ab_construction ? Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 16 [PASS] ab_gue_class ? Test 16 [MONTGOMERY] FINAL (last ?=0.5): ?r?=0.594 ? PASS

Test 16 [PASS] ab_gue_class ? Test 16b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 17 [PASS] connes_self_duality ? Test 17: zero_modes(=?=1/2)=4 (expected 4) ? PASS

Test 17 [PASS] connes_self_duality ? Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 18 [PASS] chiral_AIII ? Test 18: chiral defect (=?=1/2)=0.0 [exp 0.000000], (=?=1/3)=0.0058 [exp ?0] ? PASS

Test 18 [PASS] chiral_AIII ? Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 19 [PASS] dirac_cone ? Test 19: 1/L scaling R?=0.9997, v_F(2?)=1.8998, v_F(?)=3.7995 ? PASS

Test 19 [PASS] dirac_cone ? Test 19b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 19 [PASS] dirac_cone ? Test 19c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Test 20 [PASS] chern_tknn ? Test 20: C? anchors {1/4,1/3,1/5}=1,1,1 [exp 1,1,1], ?=1/2 min gap=0.0 (Dirac touching), mass branch C=0/0 ? PASS

Test 20 [PASS] chern_tknn ? Test 20b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 21 [PASS] gamma_phase ? Test 21: arg(?*)=89.874° (?90°, deviation=0.126°) ? PASS

Test 21 [PASS] gamma_phase ? Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 22 [PASS] ab_phase ? Test 22: $\gamma_{AB}=0.4487989505 = \pi/7$ (exact) ? PASS

Test 22 [PASS] ab_phase ? Test 22b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 23 [PASS] fractal_factor ? Test 23: $|CF_formula|=1.0$ (=1 by construction), $c_{AB}=0.02062$ (≈ 0.02063) ? PASS

Test 23 [PASS] fractal_factor ? Test 23b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 24 [PASS] dirac_string_flux ? Test 24: vortex flux OK=1/1, empty OK=5040/5040, max_empty=0.0 ? PASS

Test 24 [PASS] dirac_string_flux ? Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

Test 25 [PASS] byers_yang ? Test 25: $\gamma(q=1)=0.0$ [exp ~ 0], $\gamma(q=0.3)=0.0032$ [exp $>1e-3$] ? PASS

Test 25 [PASS] byers_yang ? Test 25b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 26 [PASS] pbc_torus ? Test 26: flux_v=-0.0 [exp -0.0], empty_ok=5040/5183, herm=OK, $\gamma_r=0.6009$? PASS

Test 26 [PASS] pbc_torus ? Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 27 [PASS] binary_chiral ? Test 27: chiral defect ($W=0$) = 0.0, ($W=1.0$) = 0.0058 ? PASS

Test 27 [PASS] binary_chiral ? Test 27b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Test 28 [FAIL] f_gue_merit ? Test 28: $f_{GUE}=0.8704$, $\gamma_{data}=0.6115$? WARN

Test 28 [FAIL] f_gue_merit ? Test 28b [HARDCORE pass 2]: 2 sub-checks, 1 failed ? WARN

Test 29 [PASS] dirac_dip ? Test 29: $\gamma(=0.5)=0.0193$ vs $\gamma(0.403)=0.1944$ / $\gamma(0.597)=0.1944$? PASS

Test 29 [FAIL] dirac_dip ? Test 29b [HARDCORE pass 2]: 3 sub-checks, 1 failed ? WARN

Test 29 [PASS] dirac_dip ? Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS

Test 30 [PASS] vf_scaling ? Test 30: $v_F=1.798$, $R=0.9977$? PASS

Test 30 [PASS] vf_scaling ? Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 31 [PASS] hatano_nelson ? Test 31: $\max|\text{Im}(E)|=0.6397$, $\gamma_{NH}=0.8938$? PASS

Test 31 [PASS] hatano_nelson ? Test 31b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Test 31 [PASS] hatano_nelson ? Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

Test 32 [PASS] rmean_bootstrap ? Test 32: $\gamma_r=0.5991-0.0075$ ($q=1.0$, $n=5$, $W=1.0$) vs GUE 0.5992 ? PASS

Test 32 [PASS] rmean_bootstrap ? Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Test 33 [PASS] l_scaling_rmean ? Test 33: L-scaling to $L=80$ 0.434,0.556,0.593,0.6,0.6 ? PASS

Test 33 [PASS] l_scaling_rmean ? Test 33b (hardcore): plateau weighted $\gamma_r=0.6004$, $\gamma_{dof}=0.01$? PASS

Test 34 [PASS] direct_vs_zeta ? Test 34: KS $p=0.0$, $\gamma_R=0.0374$, $d_{GUE}=0.796$? PASS

Test 34 [FAIL] direct_vs_zeta ? Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN

Test 35 [FAIL] form_factor_Kt ? Test 35: $K(t)$ ramp+plateau ? RMS_GUE=0.4193, corr_GUE=0.8889 ? WARN

Test 35 [FAIL] form_factor_Kt ? Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Test 36 [PASS] byte_robust ? Test 36: byte robust ? $r_{256}=0.5894$, $|r|_{\text{multi}}=0.001$ (120 windows), $|r|_{1\text{win}}=0.012$ (diag), $H=7.9991$ bits, $\gamma_z=-0.234$, $n=27997$? PASS

Test 36 [PASS] byte_robust ? Test 36: byte robust ? $r_{256}=0.5756$, $|r|_{\text{multi}}=0.0009$ (120 windows), $|r|_{1\text{win}}=0.0061$ (diag), $H=7.9991$ bits, $\gamma_z=-0.234$, $n=55288$? PASS

Test 37 [PASS] half_factorial_gamma ? Test 37: $(1/2)! = \pi/2$ [rel.err 0.0], $32/\pi$ [rel.err 0.0], $\gamma_p=1$ [$\gamma=0.0$], uniqueness ? ? PASS

Test 37 [PASS] half_factorial_gamma ? Test 37b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

Execution steps (computation-log section markers)

[13:48:04.797] [+ 0.079s] ? run_all_tests START

[13:48:04.880] [+ 0.162s] ? Test 1 ? pass 1 (primary: standard implementation)

[13:48:05.591] [+ 0.873s] pass 1 result: PASS (elapsed: 0.71s)

[13:48:05.591] [+ 0.873s] ? Test 1 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:48:27.985] [+ 23.267s] pass 2 result: PASS (elapsed: 22.39s)

[13:48:27.985] [+ 23.267s] Test 1 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:48:28.733] [+ 24.015s] ? Test 2 ? pass 1 (primary: standard implementation)

[13:48:39.425] [+ 34.707s] pass 1 result: PASS (elapsed: 10.69s)

[13:48:39.425] [+ 34.707s] ? Test 2 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:48:49.127] [+ 44.409s] pass 2 result: PASS (elapsed: 9.70s)

[13:48:49.128] [+ 44.410s] Test 2 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:48:49.817] [+ 45.099s] ? Test 3 ? pass 1 (primary: standard implementation)

[13:48:50.148] [+ 45.430s] pass 1 result: PASS (elapsed: 0.33s)

[13:48:50.148] [+ 45.430s] ? Test 3 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:48:59.449] [+ 54.732s] pass 2 result: PASS (elapsed: 9.30s)

[13:48:59.449] [+ 54.732s] Test 3 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:49:00.140] [+ 55.422s] ? Test 4 ? pass 1 (primary: standard implementation)

[13:49:00.316] [+ 55.598s] pass 1 result: FAIL/WARN (elapsed: 0.18s)

[13:49:00.316] [+ 55.598s] ? Test 4 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:49:09.999] [+ 65.282s] pass 2 result: PASS (elapsed: 9.68s)

[13:49:09.999] [+ 65.282s] Test 4 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[13:49:10.710] [+ 65.993s] ? Test 5 ? pass 1 (primary: standard implementation)

[13:49:11.198] [+ 66.480s] pass 1 result: FAIL/WARN (elapsed: 0.49s)

[13:49:11.198] [+ 66.480s] ? Test 5 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:49:21.050] [+ 76.332s] pass 2 result: PASS (elapsed: 9.85s)

[13:49:21.050] [+ 76.333s] Test 5 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[13:49:21.798] [+ 77.080s] ? Test 6 ? pass 1 (primary: standard implementation)

[13:49:22.162] [+ 77.444s] pass 1 result: FAIL/WARN (elapsed: 0.36s)

[13:49:22.162] [+ 77.444s] ? Test 6 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:49:31.754] [+ 87.036s] pass 2 result: PASS (elapsed: 9.59s)

[13:49:31.754] [+ 87.036s] Test 6 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[13:49:32.468] [+ 87.751s] ? Test 7 ? pass 1 (primary: standard implementation)

[13:49:32.709] [+ 87.991s] pass 1 result: PASS (elapsed: 0.24s)

[13:49:32.709] [+ 87.992s] ? Test 7 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:49:42.167] [+ 97.449s] pass 2 result: PASS (elapsed: 9.46s)

[13:49:42.167] [+ 97.449s] Test 7 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:49:42.913] [+ 98.196s] ? Test 8 ? pass 1 (primary: standard implementation)

[13:49:43.153] [+ 98.436s] pass 1 result: PASS (elapsed: 0.24s)

[13:49:43.154] [+ 98.436s] ? Test 8 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:49:52.662] [+107.944s] pass 2 result: PASS (elapsed: 9.51s)

[13:49:52.662] [+107.944s] Test 8 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:49:53.350] [+108.632s] ? Test 9 ? pass 1 (primary: standard implementation)

[13:49:53.580] [+108.862s] pass 1 result: PASS (elapsed: 0.23s)

[13:49:53.580] [+108.862s] ? Test 9 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:50:02.811] [+118.093s] pass 2 result: PASS (elapsed: 9.23s)

[13:50:02.811] [+118.093s] Test 9 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:50:03.524] [+118.806s] ? Test 10 ? pass 1 (primary: standard implementation)

[13:50:03.792] [+119.074s] pass 1 result: PASS (elapsed: 0.27s)

[13:50:03.792] [+119.074s] ? Test 10 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[13:50:13.786] [+129.068s] pass 2 result: PASS (elapsed: 9.99s)

[13:50:13.786] [+129.068s] Test 10 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[13:50:14.508] [+129.790s] ? Test 11 ? pass 1 (primary: standard implementation)

[14:00:20.704] [+735.986s] pass 1 result: FAIL/WARN (elapsed: 606.20s)

[14:00:20.704] [+735.986s] ? Test 11 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[14:02:24.336] [+859.618s] pass 2 result: PASS (elapsed: 123.63s)

[14:02:24.336] [+859.618s] Test 11 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[14:02:25.804] [+861.086s] ? Test 12 ? pass 1 (primary: standard implementation)

[14:02:26.074] [+861.356s] pass 1 result: FAIL/WARN (elapsed: 0.27s)

[14:02:26.074] [+861.356s] ? Test 12 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[14:02:40.228] [+875.510s] pass 2 result: PASS (elapsed: 14.15s)

[14:02:40.228] [+875.510s] Test 12 two-pass summary: pass1=FAIL pass2=PASS (hardcore) ? OVERALL FAIL

[14:02:41.217] [+876.499s] ? Test 13 ? pass 1 (primary: standard implementation)

[14:02:41.638] [+876.920s] pass 1 result: PASS (elapsed: 0.42s)

[14:02:41.638] [+876.920s] ? Test 13 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[14:02:54.599] [+889.881s] pass 2 result: PASS (elapsed: 12.96s)

[14:02:54.599] [+889.881s] Test 13 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[14:02:55.465] [+890.747s] ? Test 14 ? pass 1 (primary: standard implementation)

[14:02:56.116] [+891.398s] pass 1 result: PASS (elapsed: 0.65s)

[14:02:56.116] [+891.399s] ? Test 14 ? pass 2 (secondary: HARDCORE deep validation, no lattice dependence)

[14:03:09.104] [+904.386s] pass 2 result: PASS (elapsed: 12.99s)

[14:03:09.104] [+904.386s] Test 14 two-pass summary: pass1=PASS pass2=PASS (hardcore) ? OVERALL PASS

[14:03:09.964] [+905.246s] ? Test 15 ? pass 1 (primary: 72×72, Nv=[4])

[14:03:13.350] [+908.633s] pass 1 result: PASS (elapsed: 3.39s)

[14:03:13.351] [+908.633s] ? Test 15 ? pass 2 (secondary: 96×96, Nv=[2])

[14:03:36.815] [+932.097s] pass 2 result: PASS (elapsed: 23.46s)

[14:03:36.815] [+932.097s] Test 15 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[14:03:37.649] [+932.931s] ? Test 16 ? pass 1 (primary: 72×72, Nv=[4])

[14:03:38.066] [+933.348s] ? Test 16 single-?: ?=0.500000 (?=0.5000 (default))

[14:06:52.544] [+1127.826s] pass 1 result: PASS (elapsed: 194.89s)

[14:06:52.544] [+1127.826s] ? Test 16 ? pass 2 (secondary: 96×96, Nv=[2])

[14:32:42.865] [+2678.147s] pass 2 result: PASS (elapsed: 1550.32s)

[14:32:42.865] [+2678.147s] Test 16 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[14:32:43.885] [+2679.167s] ? Test 17 ? pass 1 (primary: 72×72, Nv=[4])

[14:40:55.746] [+3171.028s] pass 1 result: PASS (elapsed: 491.86s)

[14:40:55.747] [+3171.029s] ? Test 17 ? pass 2 (secondary: 96×96, Nv=[2])

[15:07:52.727] [+4788.010s] pass 2 result: PASS (elapsed: 1616.97s)

[15:07:52.728] [+4788.011s] Test 17 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:07:56.207] [+4791.489s] ? Test 18 ? pass 1 (primary: 72×72, Nv=[4])

[15:11:23.405] [+4998.687s] pass 1 result: PASS (elapsed: 207.20s)

[15:11:23.405] [+4998.687s] ? Test 18 ? pass 2 (secondary: 96×96, Nv=[2])

[15:12:20.943] [+5056.225s] pass 2 result: PASS (elapsed: 57.54s)

[15:12:20.943] [+5056.225s] Test 18 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:12:22.027] [+5057.309s] ? Test 19 ? pass 1 (primary: 72×72, Nv=[4])

[15:13:36.901] [+5132.184s] pass 1 result: PASS (elapsed: 74.87s)

[15:13:36.902] [+5132.184s] ? Test 19 ? pass 2 (secondary: 96×96, Nv=[2])

[15:24:25.778] [+5781.060s] pass 2 result: PASS (elapsed: 648.88s)

[15:24:25.778] [+5781.060s] ? Test 19 ? pass 3 (tertiary: SERIES ? real report series, 96×96 regime)

[15:30:16.835] [+6132.117s] ? Test 20 ? pass 1 (primary: 72×72, Nv=[4])

[15:30:17.548] [+6132.830s] pass 1 result: PASS (elapsed: 0.71s)

[15:30:17.548] [+6132.830s] ? Test 20 ? pass 2 (secondary: 96×96, Nv=[2])

[15:31:17.642] [+6192.924s] pass 2 result: PASS (elapsed: 60.09s)

[15:31:17.643] [+6192.925s] Test 20 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:31:18.615] [+6193.897s] ? Test 21 ? pass 1 (primary: 72✖2, Nv=[4])

[15:31:19.032] [+6194.314s] pass 1 result: PASS (elapsed: 0.42s)

[15:31:19.032] [+6194.314s] ? Test 21 ? pass 2 (secondary: 96✖6, Nv=[2])

[15:31:32.783] [+6208.065s] pass 2 result: PASS (elapsed: 13.75s)

[15:31:32.783] [+6208.065s] Test 21 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:31:33.746] [+6209.029s] ? Test 22 ? pass 1 (primary: 72✖2, Nv=[4])

[15:31:33.960] [+6209.243s] pass 1 result: PASS (elapsed: 0.21s)

[15:31:33.960] [+6209.243s] ? Test 22 ? pass 2 (secondary: 96✖6, Nv=[2])

[15:31:47.778] [+6223.061s] pass 2 result: PASS (elapsed: 13.82s)

[15:31:47.778] [+6223.061s] Test 22 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:31:48.745] [+6224.028s] ? Test 23 ? pass 1 (primary: 72✖2, Nv=[4])

[15:31:49.126] [+6224.408s] pass 1 result: PASS (elapsed: 0.38s)

[15:31:49.126] [+6224.408s] ? Test 23 ? pass 2 (secondary: 96✖6, Nv=[2])

[15:32:02.675] [+6237.958s] pass 2 result: PASS (elapsed: 13.55s)

[15:32:02.676] [+6237.958s] Test 23 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:32:03.588] [+6238.870s] ? Test 24 ? pass 1 (primary: 72✖2, Nv=[4])

[15:32:05.403] [+6240.686s] pass 1 result: PASS (elapsed: 1.82s)

[15:32:05.404] [+6240.686s] ? Test 24 ? pass 2 (secondary: 96✖6, Nv=[2])

[15:33:11.749] [+6307.032s] pass 2 result: PASS (elapsed: 66.35s)

[15:33:11.750] [+6307.032s] Test 24 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[15:33:12.904] [+6308.186s] ? Test 25 ? pass 1 (primary: 72✖2, Nv=[4])

[15:35:40.787] [+6456.069s] pass 1 result: PASS (elapsed: 147.88s)

[15:35:40.787] [+6456.069s] ? Test 25 ? pass 2 (secondary: 96✖6, Nv=[2])

[16:39:40.082] [+10295.364s] pass 2 result: PASS (elapsed: 3839.29s)

[16:39:40.082] [+10295.365s] Test 25 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[16:39:41.352] [+10296.634s] ? Test 26 ? pass 1 (primary: 72✖2, Nv=[4])

[16:40:40.734] [+10356.016s] pass 1 result: PASS (elapsed: 59.38s)

[16:40:40.734] [+10356.016s] ? Test 26 ? pass 2 (secondary: 96✖6, Nv=[2])

[17:09:15.511] [+12070.794s] pass 2 result: PASS (elapsed: 1714.78s)

[17:09:15.511] [+12070.794s] Test 26 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[17:09:16.249] [+12071.531s] ? Test 27 ? pass 1 (primary: 72✖2, Nv=[4])

[17:09:30.084] [+12085.366s] pass 1 result: PASS (elapsed: 13.84s)

[17:09:30.085] [+12085.367s] ? Test 27 ? pass 2 (secondary: 96✖6, Nv=[2])

[17:12:14.164] [+12249.446s] pass 2 result: PASS (elapsed: 164.08s)

[17:12:14.164] [+12249.446s] Test 27 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[17:12:15.173] [+12250.455s] ? Test 28 ? pass 1 (primary: 72✕72, Nv=[4])

[17:13:08.708] [+12303.990s] pass 1 result: FAIL/WARN (elapsed: 53.54s)

[17:13:08.708] [+12303.991s] ? Test 28 ? pass 2 (secondary: 96✕96, Nv=[2])

[17:39:36.656] [+13891.939s] pass 2 result: FAIL/WARN (elapsed: 1587.95s)

[17:39:36.657] [+13891.939s] Test 28 two-pass summary: pass1=FAIL pass2=FAIL ? OVERALL FAIL

[17:39:37.904] [+13893.186s] ? Test 29 ? pass 1 (primary: 72✕72, Nv=[4])

[17:42:15.059] [+14050.341s] pass 1 result: PASS (elapsed: 157.15s)

[17:42:15.059] [+14050.341s] ? Test 29 ? pass 2 (secondary: 96✕96, Nv=[2])

[17:57:50.396] [+14985.678s] pass 2 result: FAIL/WARN (elapsed: 935.34s)

[17:57:50.396] [+14985.678s] ? Test 29 ? pass 3 (tertiary: SERIES ? real report series, 96✕96 regime)

[18:45:06.539] [+17821.821s] ? Test 30 ? pass 1 (primary: 72✕72, Nv=[4])

[18:47:06.232] [+17941.514s] pass 1 result: PASS (elapsed: 119.69s)

[18:47:06.232] [+17941.514s] ? Test 30 ? pass 2 (secondary: 96✕96, Nv=[2])

[18:59:11.075] [+18666.357s] pass 2 result: PASS (elapsed: 724.84s)

[18:59:11.078] [+18666.360s] Test 30 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[18:59:12.283] [+18667.565s] ? Test 31 ? pass 1 (primary: 72✕72, Nv=[4])

[19:03:17.582] [+18912.864s] pass 1 result: PASS (elapsed: 245.30s)

[19:03:17.582] [+18912.864s] ? Test 31 ? pass 2 (secondary: 96✕96, Nv=[2])

[19:58:20.555] [+22215.837s] pass 2 result: PASS (elapsed: 3302.97s)

[19:58:20.555] [+22215.837s] ? Test 31 ? pass 3 (tertiary: SERIES ? real report series, 96✕96 regime)

[20:04:51.687] [+22606.969s] ? Test 32 ? pass 1 (primary: 72✕72, Nv=[4])

[20:13:27.441] [+23122.723s] pass 1 result: PASS (elapsed: 515.75s)

[20:13:27.441] [+23122.723s] ? Test 32 ? pass 2 (secondary: 96✕96, Nv=[2])

[21:02:32.804] [+26068.086s] pass 2 result: PASS (elapsed: 2945.36s)

[21:02:32.804] [+26068.086s] Test 32 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[21:02:33.557] [+26068.839s] ? Test 33 ? pass 1 (primary: 72✕72, Nv=[4])

[21:18:17.750] [+27013.032s] pass 1 result: PASS (elapsed: 944.19s)

[21:18:17.750] [+27013.032s] ? Test 33 ? pass 2 (secondary: 96✕96, Nv=[2])

[21:33:01.262] [+27896.544s] pass 2 result: PASS (elapsed: 883.51s)

[21:33:01.262] [+27896.544s] Test 33 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[21:33:02.273] [+27897.555s] ? Test 34 ? pass 1 (primary: 72✕72, Nv=[4])

[21:37:02.924] [+28138.206s] pass 1 result: PASS (elapsed: 240.65s)

[21:37:02.924] [+28138.206s] ? Test 34 ? pass 2 (secondary: 96✖06, Nv=[2])

[22:04:23.597] [+29778.879s] pass 2 result: FAIL/WARN (elapsed: 1640.67s)

[22:04:23.600] [+29778.882s] Test 34 two-pass summary: pass1=PASS pass2=FAIL ? OVERALL FAIL

[22:04:24.906] [+29780.188s] ? Test 35 ? pass 1 (primary: 72✖72, Nv=[4])

[22:09:08.652] [+30063.934s] pass 1 result: FAIL/WARN (elapsed: 283.74s)

[22:09:08.653] [+30063.936s] ? Test 35 ? pass 2 (secondary: 96✖06, Nv=[2])

[22:37:01.653] [+31736.935s] pass 2 result: FAIL/WARN (elapsed: 1673.00s)

[22:37:01.654] [+31736.936s] Test 35 two-pass summary: pass1=FAIL pass2=FAIL ? OVERALL FAIL

[22:37:02.779] [+31738.061s] ? Test 36 ? pass 1 (primary: 72✖72, Nv=[4])

[22:45:52.588] [+32267.870s] pass 1 result: PASS (elapsed: 529.81s)

[22:45:52.588] [+32267.870s] ? Test 36 ? pass 2 (secondary: 96✖06, Nv=[2])

[23:33:00.768] [+35096.050s] pass 2 result: PASS (elapsed: 2828.18s)

[23:33:00.768] [+35096.050s] Test 36 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

[23:33:02.097] [+35097.379s] ? Test 37 ? pass 1 (primary: 72✖72, Nv=[4])

[23:33:21.765] [+35117.048s] pass 1 result: PASS (elapsed: 19.67s)

[23:33:21.766] [+35117.048s] ? Test 37 ? pass 2 (secondary: 96✖06, Nv=[2])

[23:33:39.517] [+35134.799s] pass 2 result: PASS (elapsed: 17.75s)

[23:33:39.517] [+35134.799s] Test 37 two-pass summary: pass1=PASS pass2=PASS ? OVERALL PASS

test_01_bN_convergence [PASS] reports/ + logs/ copied; plots live in ../test_01_bN_convergence/plots/

test_01_bN_convergence [PASS] reports/ + logs/ copied; plots live in ../test_01_bN_convergence/plots/

test_02_bN_monotonicity [PASS] reports/ + logs/ copied; plots live in ../test_02_bN_monotonicity/plots/

test_02_bN_monotonicity [PASS] reports/ + logs/ copied; plots live in ../test_02_bN_monotonicity/plots/

test_03_bN_rate [PASS] reports/ + logs/ copied; plots live in ../test_03_bN_rate/plots/

test_03_bN_rate [PASS] reports/ + logs/ copied; plots live in ../test_03_bN_rate/plots/

test_04_gue_ks_full [FAIL] reports/ + logs/ copied; plots live in ../test_04_gue_ks_full/plots/

test_04_gue_ks_full [PASS] reports/ + logs/ copied; plots live in ../test_04_gue_ks_full/plots/

test_05_gue_ks_highT [FAIL] reports/ + logs/ copied; plots live in ../test_05_gue_ks_highT/plots/

test_05_gue_ks_highT [PASS] reports/ + logs/ copied; plots live in ../test_05_gue_ks_highT/plots/

test_06_chi2_hist [FAIL] reports/ + logs/ copied; plots live in ../test_06_chi2_hist/plots/

test_06_chi2_hist [PASS] reports/ + logs/ copied; plots live in ../test_06_chi2_hist/plots/

test_07_decay_slope [PASS] reports/ + logs/ copied; plots live in ../test_07_decay_slope/plots/

test_07_decay_slope [PASS] reports/ + logs/ copied; plots live in ../test_07_decay_slope/plots/

test_08_residuals [PASS] reports/ + logs/ copied; plots live in ../test_08_residuals/plots/

Artifact map

test_08_residuals [PASS] reports/ + logs/ copied; plots live in ../test_08_residuals/plots/

test_09_bootstrap_ci [PASS] reports/ + logs/ copied; plots live in ../test_09_bootstrap_ci/plots/

test_09_bootstrap_ci [PASS] reports/ + logs/ copied; plots live in ../test_09_bootstrap_ci/plots/

test_10_cross_validation [PASS] reports/ + logs/ copied; plots live in ../test_10_cross_validation/plots/

test_10_cross_validation [PASS] reports/ + logs/ copied; plots live in ../test_10_cross_validation/plots/

test_11_anderson_darling [FAIL] reports/ + logs/ copied; plots live in ../test_11_anderson_darling/plots/

test_11_anderson_darling [PASS] reports/ + logs/ copied; plots live in ../test_11_anderson_darling/plots/

test_12_two_sample_ks [FAIL] reports/ + logs/ copied; plots live in ../test_12_two_sample_ks/plots/

test_12_two_sample_ks [PASS] reports/ + logs/ copied; plots live in ../test_12_two_sample_ks/plots/

test_13_number_variance [PASS] reports/ + logs/ copied; plots live in ../test_13_number_variance/plots/

test_13_number_variance [PASS] reports/ + logs/ copied; plots live in ../test_13_number_variance/plots/

test_14_spectral_rigidity [PASS] reports/ + logs/ copied; plots live in ../test_14_spectral_rigidity/plots/

test_14_spectral_rigidity [PASS] reports/ + logs/ copied; plots live in ../test_14_spectral_rigidity/plots/

test_15_ab_construction [PASS] reports/ + logs/ copied; plots live in ../test_15_ab_construction/plots/

test_15_ab_construction [PASS] reports/ + logs/ copied; plots live in ../test_15_ab_construction/plots/

test_16_ab_gue_class [PASS] reports/ + logs/ copied; plots live in ../test_16_ab_gue_class/plots/

test_16_ab_gue_class [PASS] reports/ + logs/ copied; plots live in ../test_16_ab_gue_class/plots/

test_17_connes_self_duality [PASS] reports/ + logs/ copied; plots live in ../test_17_connes_self_duality/plots/

test_17_connes_self_duality [PASS] reports/ + logs/ copied; plots live in ../test_17_connes_self_duality/plots/

test_18_chiral_AIII [PASS] reports/ + logs/ copied; plots live in ../test_18_chiral_AIII/plots/

test_18_chiral_AIII [PASS] reports/ + logs/ copied; plots live in ../test_18_chiral_AIII/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_19_dirac_cone [PASS] reports/ + logs/ copied; plots live in ../test_19_dirac_cone/plots/

test_20_chern_tknn [PASS] reports/ + logs/ copied; plots live in ../test_20_chern_tknn/plots/

test_20_chern_tknn [PASS] reports/ + logs/ copied; plots live in ../test_20_chern_tknn/plots/

test_21_gamma_phase [PASS] reports/ + logs/ copied; plots live in ../test_21_gamma_phase/plots/

test_21_gamma_phase [PASS] reports/ + logs/ copied; plots live in ../test_21_gamma_phase/plots/

test_22_ab_phase [PASS] reports/ + logs/ copied; plots live in ../test_22_ab_phase/plots/

test_22_ab_phase [PASS] reports/ + logs/ copied; plots live in ../test_22_ab_phase/plots/

test_23_fractal_factor [PASS] reports/ + logs/ copied; plots live in ../test_23_fractal_factor/plots/

test_23_fractal_factor [PASS] reports/ + logs/ copied; plots live in ../test_23_fractal_factor/plots/

test_24_dirac_string_flux [PASS] reports/ + logs/ copied; plots live in ../test_24_dirac_string_flux/plots/

test_24_dirac_string_flux [PASS] reports/ + logs/ copied; plots live in ../test_24_dirac_string_flux/plots/

test_25_byers_yang [PASS] reports/ + logs/ copied; plots live in ../test_25_byers_yang/plots/

test_25_byers_yang [PASS] reports/ + logs/ copied; plots live in ../test_25_byers_yang/plots/

test_26_pbc_torus [PASS] reports/ + logs/ copied; plots live in ../test_26_pbc_torus/plots/

test_26_pbc_torus [PASS] reports/ + logs/ copied; plots live in ../test_26_pbc_torus/plots/

test_27_binary_chiral [PASS] reports/ + logs/ copied; plots live in ../test_27_binary_chiral/plots/

test_27_binary_chiral [PASS] reports/ + logs/ copied; plots live in ../test_27_binary_chiral/plots/

test_28_f_gue_merit [FAIL] reports/ + logs/ copied; plots live in ../test_28_f_gue_merit/plots/

test_28_f_gue_merit [FAIL] reports/ + logs/ copied; plots live in ../test_28_f_gue_merit/plots/

test_29_dirac_dip [PASS] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_29_dirac_dip [FAIL] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_29_dirac_dip [PASS] reports/ + logs/ copied; plots live in ../test_29_dirac_dip/plots/

test_30_vf_scaling [PASS] reports/ + logs/ copied; plots live in ../test_30_vf_scaling/plots/

test_30_vf_scaling [PASS] reports/ + logs/ copied; plots live in ../test_30_vf_scaling/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_31_hatano_nelson [PASS] reports/ + logs/ copied; plots live in ../test_31_hatano_nelson/plots/

test_32_rmean_bootstrap [PASS] reports/ + logs/ copied; plots live in ../test_32_rmean_bootstrap/plots/

test_32_rmean_bootstrap [PASS] reports/ + logs/ copied; plots live in ../test_32_rmean_bootstrap/plots/

test_33_l_scaling_rmean [PASS] reports/ + logs/ copied; plots live in ../test_33_l_scaling_rmean/plots/

test_33_l_scaling_rmean [PASS] reports/ + logs/ copied; plots live in ../test_33_l_scaling_rmean/plots/

test_34_direct_vs_zeta [PASS] reports/ + logs/ copied; plots live in ../test_34_direct_vs_zeta/plots/

test_34_direct_vs_zeta [FAIL] reports/ + logs/ copied; plots live in ../test_34_direct_vs_zeta/plots/

test_35_form_factor_Kt [FAIL] reports/ + logs/ copied; plots live in ../test_35_form_factor_Kt/plots/

test_35_form_factor_Kt [FAIL] reports/ + logs/ copied; plots live in ../test_35_form_factor_Kt/plots/

test_36_byte_robust [PASS] reports/ + logs/ copied; plots live in ../test_36_byte_robust/plots/

test_36_byte_robust [PASS] reports/ + logs/ copied; plots live in ../test_36_byte_robust/plots/

test_37_half_factorial_gamma [PASS] reports/ + logs/ copied; plots live in ../test_37_half_factorial_gamma/plots/

test_37_half_factorial_gamma [PASS] reports/ + logs/ copied; plots live in ../test_37_half_factorial_gamma/plots/

Log files: logs/full_execution_log.txt (console tee, suite runs), logs/computation_log.txt / computation_log_full.txt (every computation), logs/results_verdicts.txt (one-line verdicts), log

Run index: ../index.html ? verdict table with direct links to every per-test report.